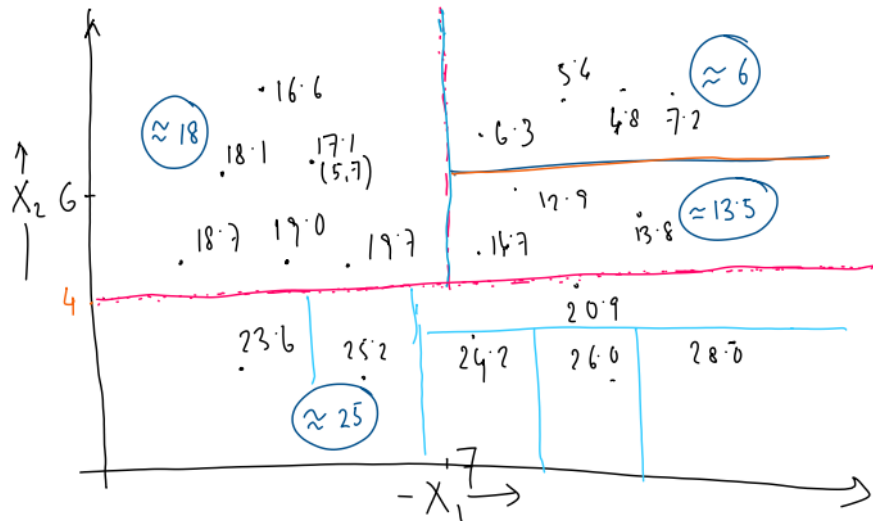


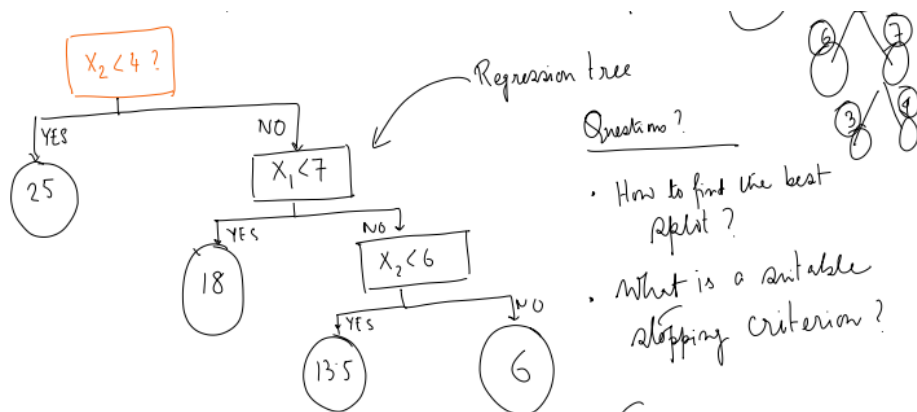
Lecture 9

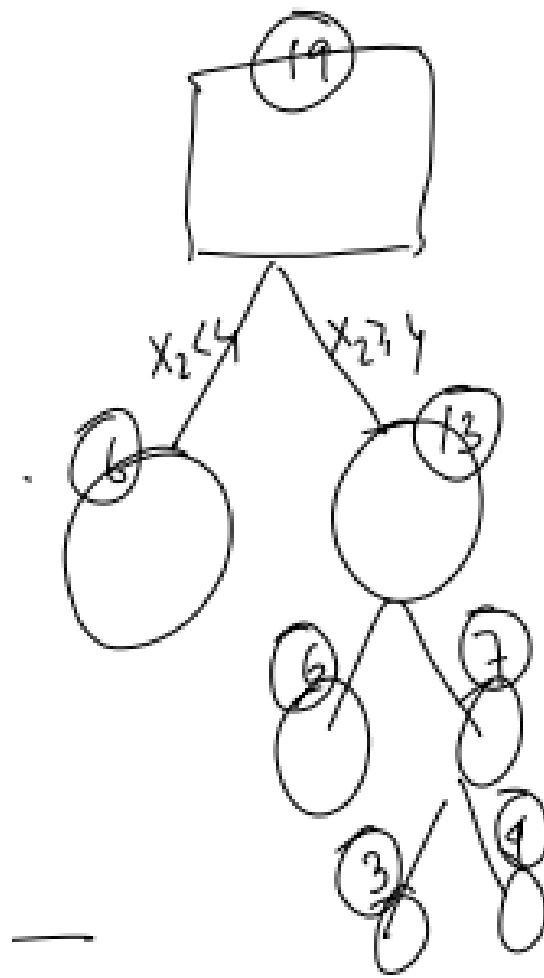
Piecewise constant estimate of a regression function where the pieces will be formed adaptively



Dataset example

X_1	X_2	Y
5	7	17.1
$\vdots$	$\vdots$	$\vdots$



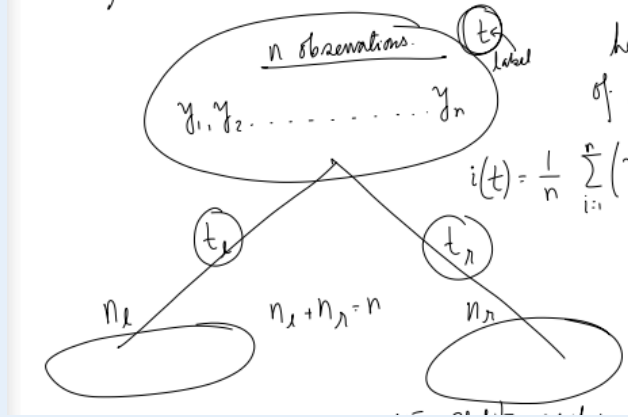


Questions:

- How to find the best split?
- What is a suitable stopping criterion?

Here at each stage, one should try to find the split such that the observations in each of the newly formed sub-regions are more homogeneous compared to the observations in the whole region.

The heterogeneity of a node can be calculated as the variance of the observations (var of the Y -values) belonging to that node.



Heterogeneity (impurity) of the node t

$$i(t) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_t)^2, \quad \text{where } \bar{y}_t = \frac{1}{n} \sum_{i=1}^n y_i$$

For the left and right child nodes:

$$i(t_L) = \frac{1}{n_L} \sum_{i: \text{obs. belongs to } t_L} (y_i - \bar{y}_{t_L})^2$$

$$i(t_R) = \frac{1}{n_R} \sum_{i: \text{obs. belongs to } t_R} (y_i - \bar{y}_{t_R})^2$$

Ideally one should choose the split so that $i(t_L)$ and $i(t_R)$ are much smaller than $i(t)$.

We define the **reduction in impurity**

$$\Delta i(t) = i(t) - \left\{ \frac{n_L}{n} i(t_L) + \frac{n_R}{n} i(t_R) \right\}.$$

Naturally, one would prefer the split for which the reduction is maximum.

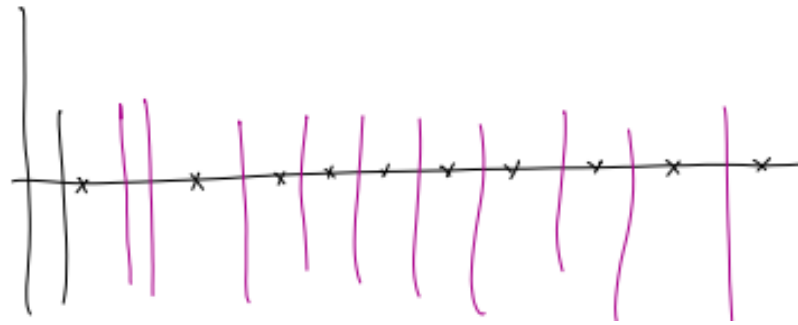
Result: For any split of the node t , $\Delta i(t) \geq 0$.

$$\begin{aligned} nS^2 &= n_L S_L^2 + n_R S_R^2 + n_L d_L^2 + n_R d_R^2 \\ &= n_L S_L^2 + n_R S_R^2 + n_L (\bar{y}_L - \bar{y})^2 + n_R (\bar{y}_R - \bar{y})^2 \end{aligned}$$

$$ni(t) = n_L i(t_L) + n_R i(t_R) + n_L (\bar{y}_L - \bar{y})^2 + n_R (\bar{y}_R - \bar{y})^2$$

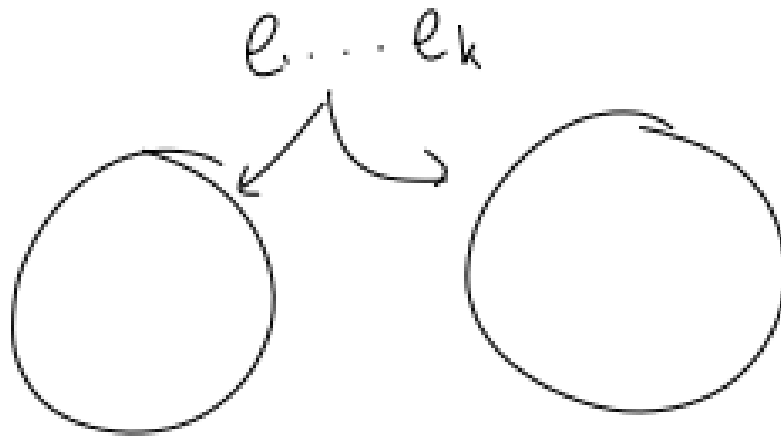
$$i(t) - \left\{ \frac{n_L}{n} i(t_L) + \frac{n_R}{n} i(t_R) \right\} = \frac{n_L}{n} (\bar{y}_L - \bar{y})^2 + \frac{n_R}{n} (\bar{y}_R - \bar{y})^2$$

$$\approx \frac{n_L n_R}{n^2} (\bar{y}_L - \bar{y}_R)^2 \geq 0.$$



If X_1 is a continuous r.v., then there are $n - 1$ many possible splits. We can calculate $\Delta i(t)$ for each of them and consider the best one.

(Discrete-valued var or ordinal var can be handled in the same way. There the no. of candidate splits = # of distinct values - 1.)



If the predictor variable is categorical (suppose it has K categories $e_1, e_2, \dots, e_K$), then we have

$$\frac{2^K - 2}{2}$$

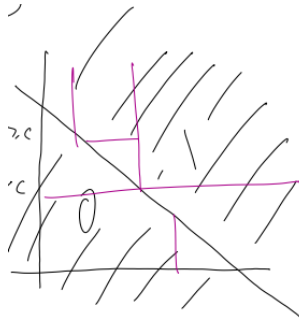
possible candidate splits and one needs to choose the best one.

After finding the best split based on each of the predictors, we finally choose the one which gives the maximum reduction.

Advantages

- It can handle several types of variables.
- Can be conveniently used for high-dimensional data.
- Several splits may be needed to approximate a simple function

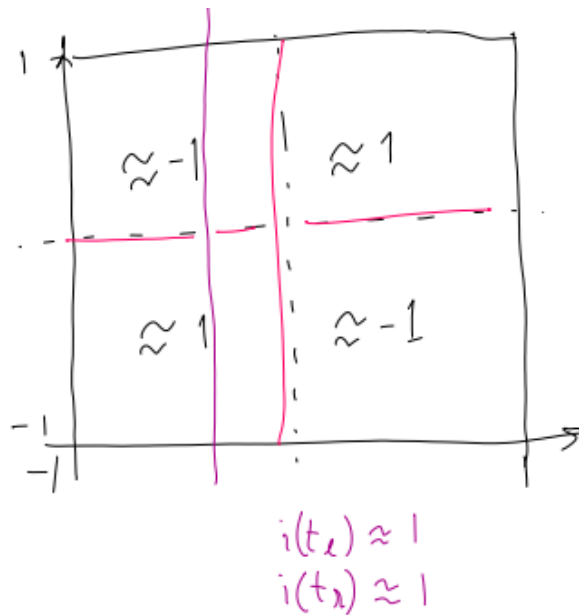
$$f(x) = \begin{cases} 1 & \text{if } aX_1 + bx_2 \geq c, \\ 0 & \text{if } aX_1 + bx_2 < c. \end{cases}$$



- The regression tree estimate usually does have high variance (a slight change in the data may lead to an altogether different type of tree).

When to stop?

- $\Delta i(t) < \varepsilon$. We can define threshold ε .
- The no. of obs. in a region (node) is less than a threshold (say 5, or 1% of the training data).
- If $i(t) < \varepsilon_0$, we may not go for further splitting.



Illustrative example

$$X_1, X_2 \sim U[-1, 1] \times U[-1, 1], \quad Y = f(X_1, X_2) + \varepsilon, \quad \varepsilon \sim N(0, 0.01)$$

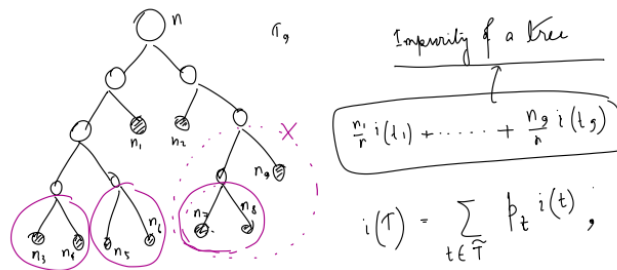
$$f(x_1, x_2) = \begin{cases} +1 & \text{if } x_1 x_2 > 0, \\ -1 & \text{if } x_1 x_2 \leq 0. \end{cases}$$

In this case $i(t) \approx 1$, $i(t_L) \approx 1$, $i(t_R) \approx 1$.

Lecture 10

Pruning

After growing a large-sized tree (using an appropriate stopping criterion), we usually do pruning to get smaller trees.



Impurity of a tree

$$i(T) = \sum_{t \in \tilde{T}} p_t i(t),$$

where $\tilde{T}$ denotes the collection of leaf nodes in T , and

p_t = proportion of observations in the node t .

During pruning, at every stage, we merge two leaf nodes so that the resulting tree will have one leaf node less than the previous one.

We always choose those two nodes whose merging leads to the least increment in the impurity of the tree.

In this way, we get a nested sequence of trees indexed by the no. of leaf nodes

$$T_M \succ T_{M-1} \succ T_{M-2} \succ \cdots \succ T_2 \succ T_1 \text{ (no split).}$$

BFOS (1984)

Breiman, Friedman, Olshen, Stone (1984) proposed to use cost-complexity & cross-validation to choose one of these models and hence the right-sized tree.

They proposed to use a cost function

$$R_\alpha(T) = R(T) + \alpha |\tilde{T}|,$$

where

- $R(T)$ = residual sum of squares,
- $|\tilde{T}|$ = no. of leaf nodes in T ,
- $\alpha > 0$ is the penalty parameter.

Choice of α	Best model
$(0, \alpha_0)$	T_M
(α_0, α_1)	T_{M-1}
(α_1, α_2)	T_{M-2}
$\vdots$	$\vdots$
(α_n, ∞)	T_1 (no split)

$$\begin{array}{c|c}
 \underline{T_M} & \underline{T_k} \quad (k < M) \\
 R(T_M) & R(T_k) \\
 + \alpha M & + \alpha k
 \end{array}$$

T_k is better than T_M if

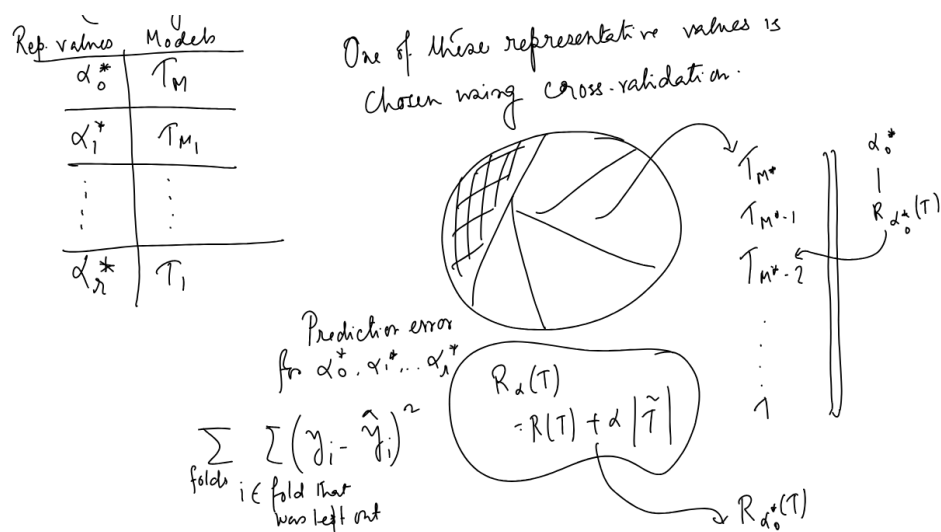
$$R(T_M) + \alpha M > R(T_k) + \alpha k \quad \Rightarrow \quad \alpha > \frac{R(T_k) - R(T_M)}{M - k}.$$

$$\alpha_0 = \min_{1 \leq k < M} \frac{R(T_k) - R(T_M)}{M - k}.$$

This process automatically deletes some of the candidates.

For each of these intervals, we choose a representative value (usually AM/GM of the upper & lower limits).

One of these representative values is chosen using cross-validation.



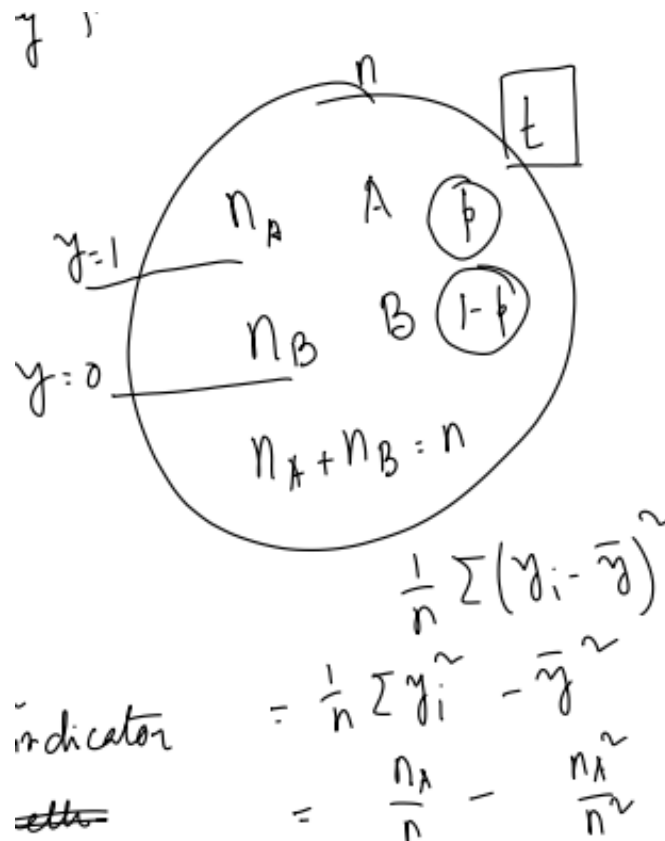
Prediction error for $\alpha_0^*, \alpha_1^*, \dots, \alpha_n^*$ is estimated by

$$\sum_{\text{folds } i \in \text{fold that was left out}} \sum (y_i - \hat{y}_i)^2$$

and the corresponding $R_{\alpha^*}(T)$ is selected.

Classification tree (Decision tree)

- We need to choose an appropriate impurity function.
- How to classify a future observation.



If we use the regression tree approach and use the variance as the impurity function, we get

$$i(t) = \frac{n_A n_B}{n^2} = p_A p_B = p(1-p).$$

(For binary classification, one can use an indicator variable and follow the regression tree algorithm for constructing the tree.)

A region corresponding to a leaf node t is assigned as a favourable region for class-1 or class-2 depending on whether $\bar{y}_t > 0.5$ or $\bar{y}_t \leq 0.5$.

What if there are more than two classes?

$$i(t) = \psi(p_1, p_2, \dots, p_K), \quad \sum_{i=1}^K p_i = 1.$$

Desirable properties

1. ψ should be symmetric in its arguments:

$$\psi(p_1, \dots, p_K) = \psi(p_{\pi(1)}, \dots, p_{\pi(K)})$$

for any permutation π of $\{1, \dots, K\}$.

2. ψ should be maximized when $p_i = 1/K$ for $i = 1, 2, \dots, K$.
3. ψ should be minimized when $p_i = 1$ for one i and $p_j = 0$ for the rest.

... ..

$p_i = 1$ for one, value of i and
 $p_i = 0$ for the rest.

split

$$\sum_{i=1}^K \tilde{p}_i = 1$$

$$\sum \tilde{p}_i \leq \sum p_i = 1$$

4. ψ should be such that the reduction in impurity for any split is non-negative.

If $\psi(p_1, \dots, p_K)$ is symmetric in its arguments and it is concave, it will satisfy all four properties mentioned above.

End of notes for W05